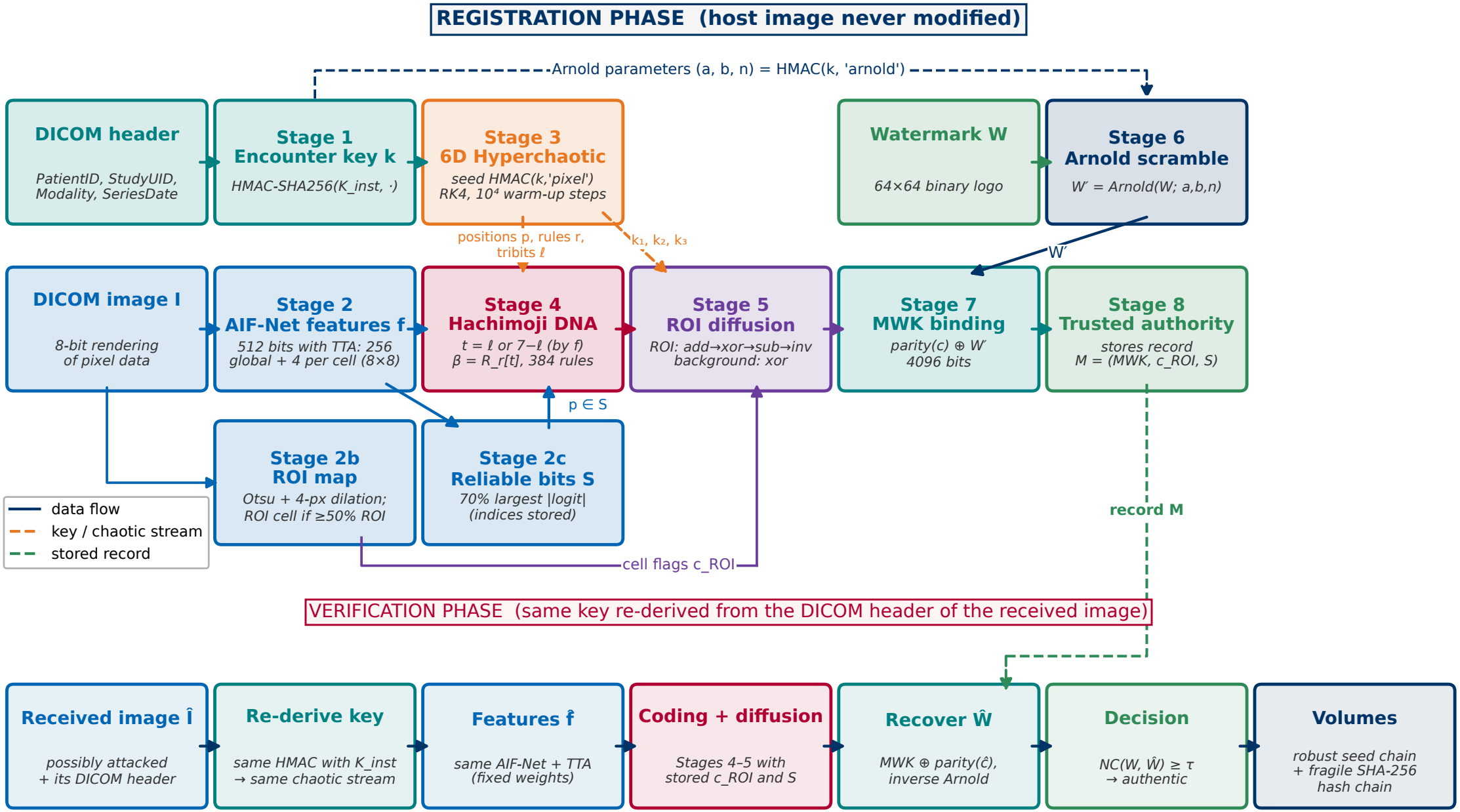

REGISTRATION PHASE (host image never modified)
Arnold parameters (a, b, n) = HMAC(k, 'arnold')
DICOM header
PatientID, StudyUID, Modality, SeriesDate
Stage 1 Encounter key k
HMAC-SHA256(K_inst, ·)
Stage 3 6D Hyperchaotic
seed HMAC(k,'pixel') RK4, 10⁴ warm-up steps
Watermark W
64×64 binary logo
Stage 6 Arnold scramble
W′ = Arnold(W; a,b,n)
positions p, rules r, tribits ℓ
k₁, k₂, k₃
W′
DICOM image I
8-bit rendering of pixel data
Stage 2 AIF-Net features f
512 bits with TTA: 256 global + 4 per cell (8×8)
Stage 4 Hachimoji DNA
t = ℓ or 7−ℓ (by f) β = R_r[t], 384 rules
Stage 5 ROI diffusion
ROI: add→xor→sub→inv background: xor
Stage 7 MWK binding
parity(c) ⊕ W′ 4096 bits
Stage 8 Trusted authority
stores record M = (MWK, c_ROI, S)
p ∈ S
Stage 2b ROI map
Otsu + 4-px dilation; ROI cell if ≥50% ROI
Stage 2c Reliable bits S
70% largest |logit| (indices stored)
data flow
key / chaotic stream
stored record
record M
cell flags c_ROI
VERIFICATION PHASE (same key re-derived from the DICOM header of the received image)
Received image Î
possibly attacked + its DICOM header
Re-derive key
same HMAC with K_inst → same chaotic stream
Features f̂
same AIF-Net + TTA (fixed weights)
Coding + diffusion
Stages 4–5 with stored c_ROI and S
Recover Ŵ
MWK ⊕ parity(ĉ), inverse Arnold
Decision
NC(W, Ŵ) ≥ τ → authentic
Volumes
robust seed chain + fragile SHA-256 hash chain